AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 5 · STUDENT LESSON

Pictures Become Numbers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Keep the positions, not just the count.

Agree on an image code, reconstruct a grid, compare two badges and explain what a pixel count loses.

Keep one habit: State the dimensions and convention → preserve each position → separate reconstruction from interpretation.

MAIN LESSON · SECTION 1

How can numbers preserve a picture?

> Keep the pixel values, their positions and the agreed code together.

Start with an incomplete message

A friend sends you six numbers: 1,0,1,0,1,0. Is that one row, two rows or three? The numbers need a shared map of positions before you can reconstruct the intended picture.

Retrieve the fitting limit

Two thresholds can agree on all **fitting** examples but disagree on a new **input**. Does zero **fitting** error identify one uniquely correct rule?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can encode and reconstruct a small image using an explicit **grid** convention.
- I can explain why **pixel** counts alone cannot recover an arrangement.
- I can distinguish numerical image data from a model's **interpretation** of that data.

▼ Earlier wording, alternative practice and picture transcripts

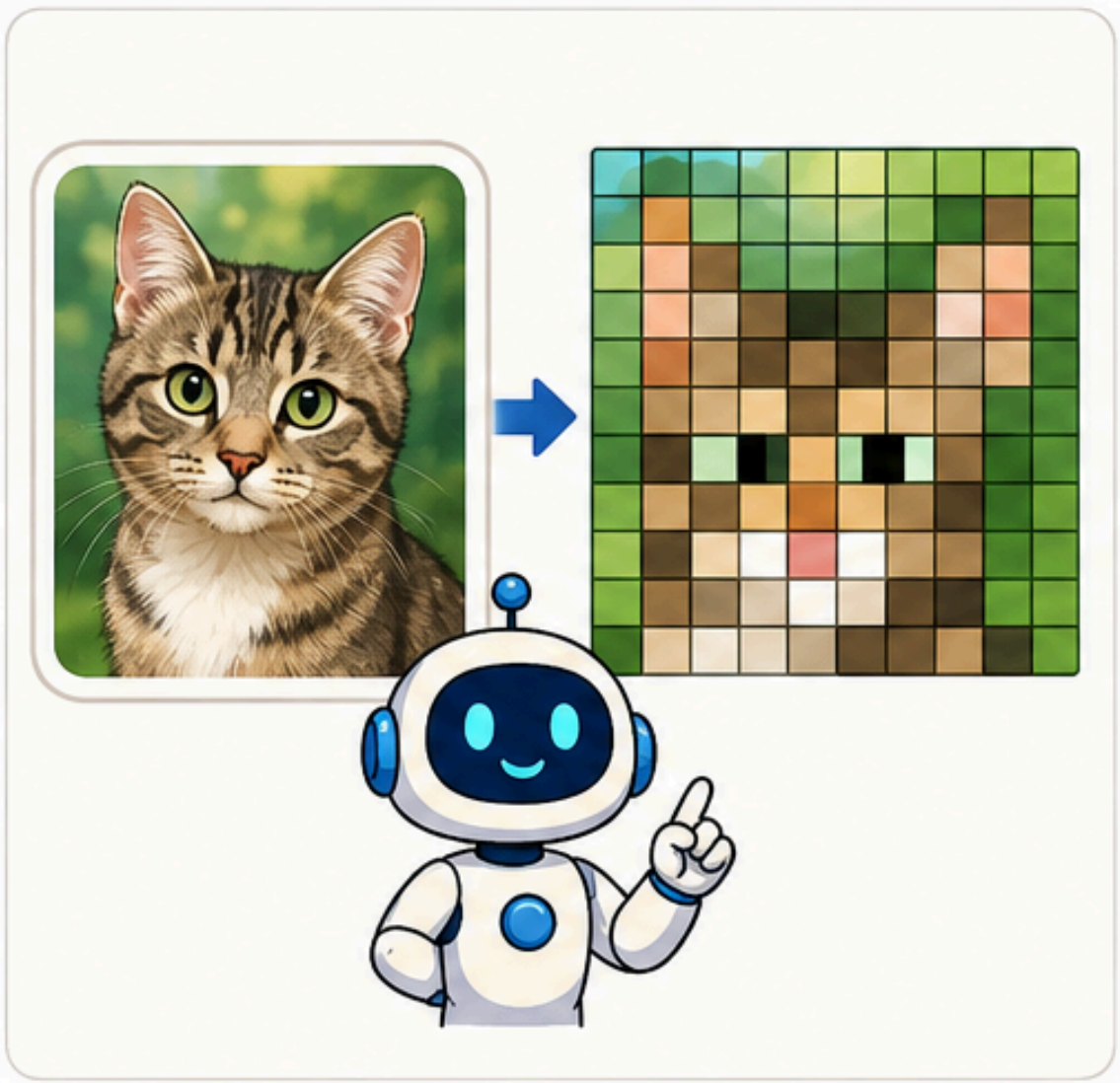
Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

A digital image is a **grid** of measured color values.



LEARNING TARGET Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first.

Explain why the image changes although the six values are identical.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can encode **pixel** positions and show why a count cannot reconstruct a picture.

MAIN LESSON · SECTION 2

Agree on the code before drawing

> Our exercise uses 1 for black and 0 for white, with rows read from the top and left to right.

Send a picture using numbers

Imagine sending a small black-and-white sign using only numbers. We can divide the sign into a **grid**, record 1 for a black square and 0 for a white square, and read each row from left to right, starting at the top.

The receiver needs the width, height and these conventions to reconstruct the same arrangement.

A convention belongs to a particular format

A **pixel** is a position with a recorded value. In our simplified binary image, one value selects black or white. Other image formats can record brightness levels or multiple colour **channel**s.

The convention 1 means black belongs to this exercise; it is not a universal rule for every digital image.

Storage is different from recognition

Encoding a picture does not recognise what it depicts. A program can store or display the numbers without calling the image a letter, an arrow or a person.

Interpretation is another operation, just as displaying a book title was different from reading the entire book in Week 2.

Check the message instructions

- Before decoding, tell your partner the width, height, reading order and what 0 and 1 mean. Which of these facts was missing from the opening six-number message?

▼ Earlier wording, alternative practice and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each **pixel** position.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Decode the grid, then read it back

> Check every position rather than guessing the shape from a familiar-looking pattern.

Read the three-by-three code

A three by three binary image

Read rows top to bottom. Within each row read left to right. 1 is black and 0 is white.

0	1	0
1	1	1
0	1	0

The image has width 3 and height 3, so it contains 9 pixels. Five are black. The row records preserve where those five black pixels occur.

Explain each row

The first row is 0,1,0, so only its middle **pixel** is black. The second row is 1,1,1, so all three are black. The third row repeats the first. This reconstructs the displayed cross-like pattern under our convention.

Keep the shape with the flattened list

A flattened record would be 0,1,0,1,1,1,0,1,0. Without the **grid** dimensions, the same nine values could also be drawn in one row of nine. Keep the shape information with the values.

Width times height counts **pixel**s; it does not tell us their contents.

Connect a sentence to a pixel

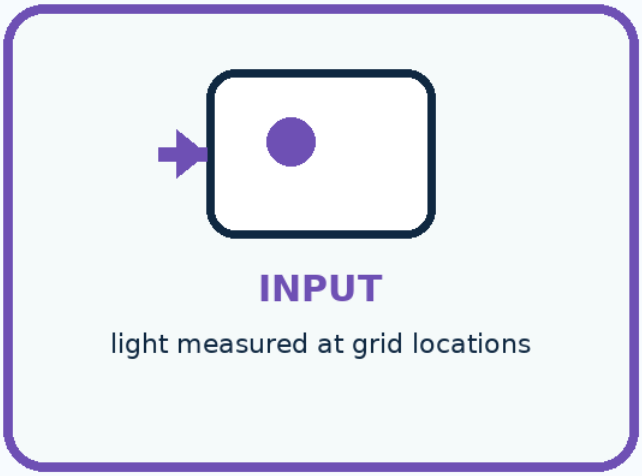
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

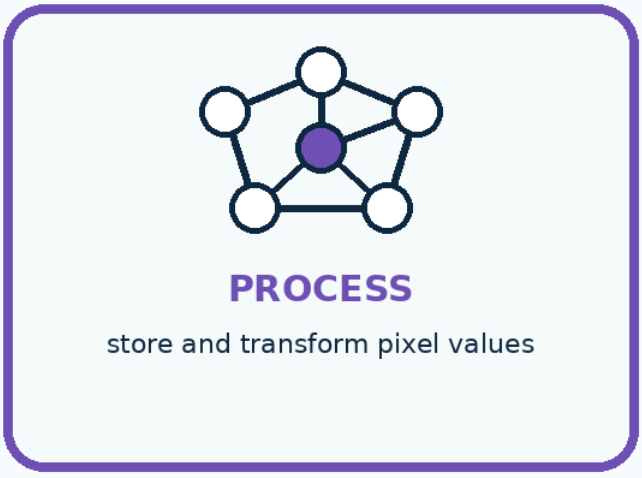
▼ Earlier wording, alternative practice and picture transcripts

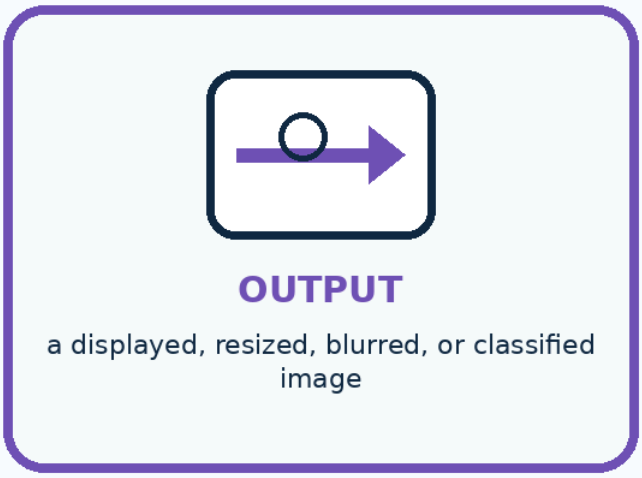
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT light measured at **grid** locations

PROCESS store and transform **pixel** values

OUTPUT a displayed, resized, blurred, or classified image

CORE IDEA A digital image is a **grid** of measured color values.

UNDER THE HOOD Digital images sample a scene across a spatial **grid**. Each **pixel** stores **channel** values, commonly red, green, and blue.

Resolution controls sampling density; transformations operate on numbers, not on an intact object inside the computer.

Draw a fresh example with all three stages:

A closer explanation

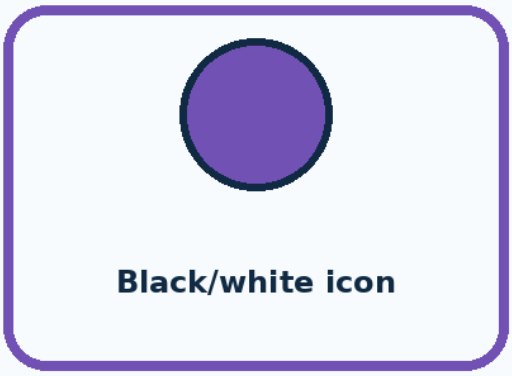
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare the image clues

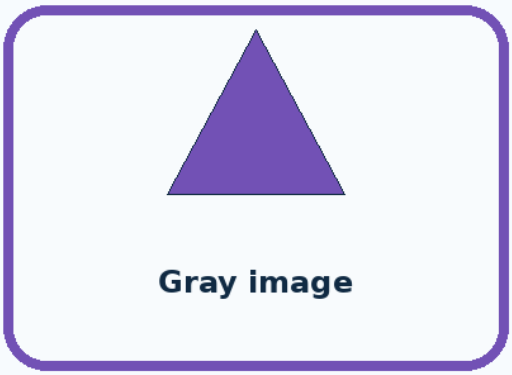
> Use the supplied picture and caption together to distinguish a measurement from an interpretation.

Four cases, paired with their explanations



The case to investigate

CASE 1 — Black/white icon Clue: 0 and 1 values. Meaning: two possible brightness levels.



The case to investigate

CASE 2 — Gray image Clue: one brightness number per **pixel**. Meaning: many light levels.



The case to investigate

CASE 3 — Color image Clue: red, green, blue values. Meaning: three **channel**s.



The case to investigate

CASE 4 — Blur Clue: nearby values are mixed. Meaning: detail is reduced.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the pixel boundary clear

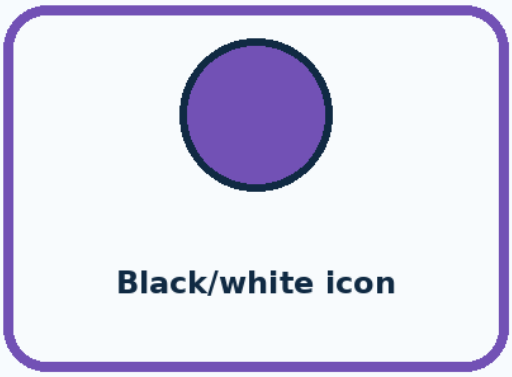
BOUNDARY A **pixel** is a measurement at a location, not a tiny copy of the object.

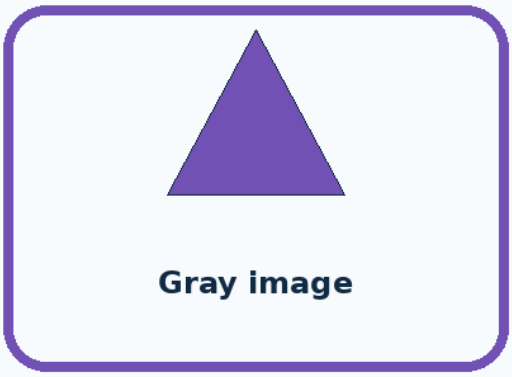
▼ Earlier wording, alternative practice and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Black/white icon Clue: 0 and 1 values. Meaning: two possible brightness levels.

CASE 2 — Gray image Clue: one brightness number per **pixel**. Meaning: many light levels.

CASE 3 — Color image Clue: red, green, blue values. Meaning: three **channel**s.

CASE 4 — Blur Clue: nearby values are mixed. Meaning: detail is reduced.

Change one thing in the pictured case

Change: Change A's bottom-left **pixel** from 0 to 1.

Prediction: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Two badges. Three dark pixels. Different pictures.

> A count preserves how many, while a grid preserves where.

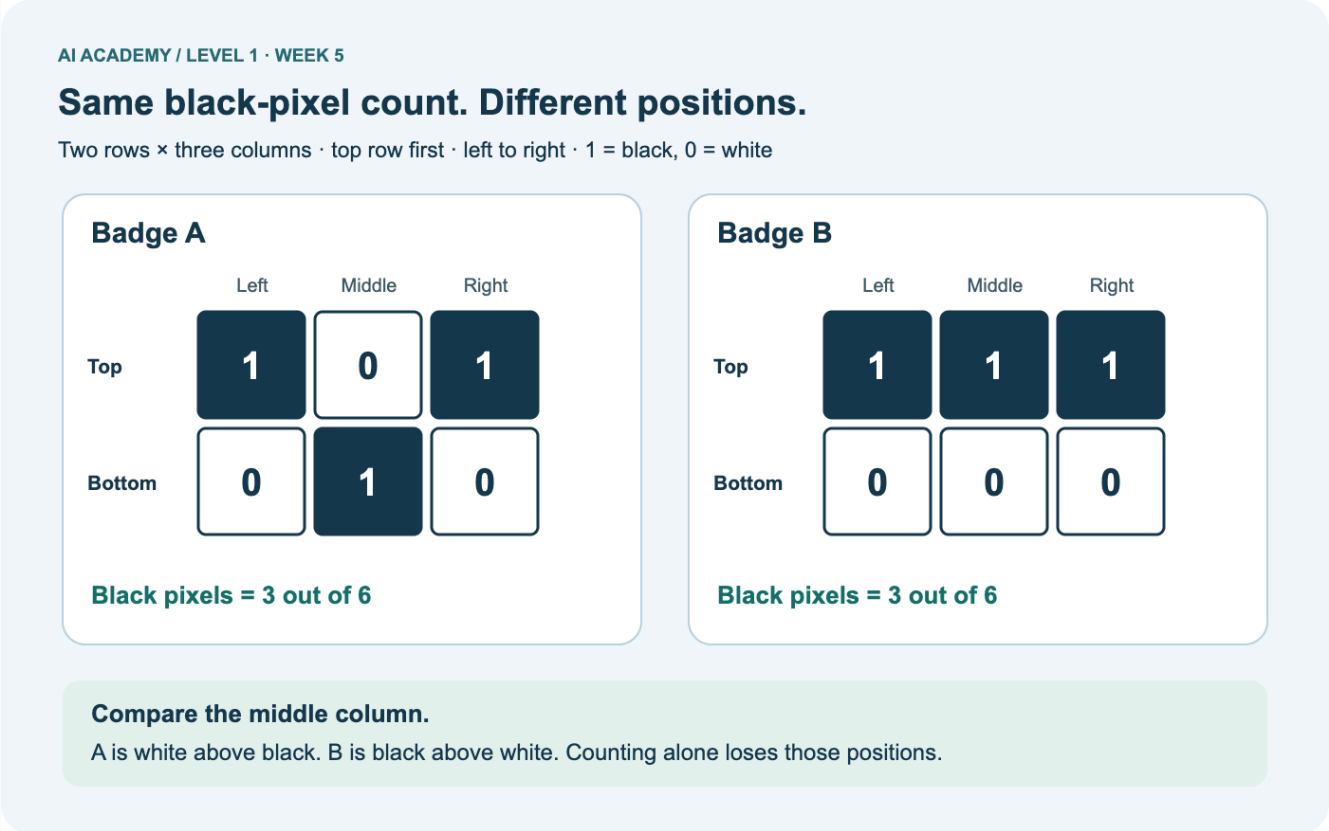
The supplied badge convention and rows

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

Dark=1, light=0; rows are read top to bottom, left to right.

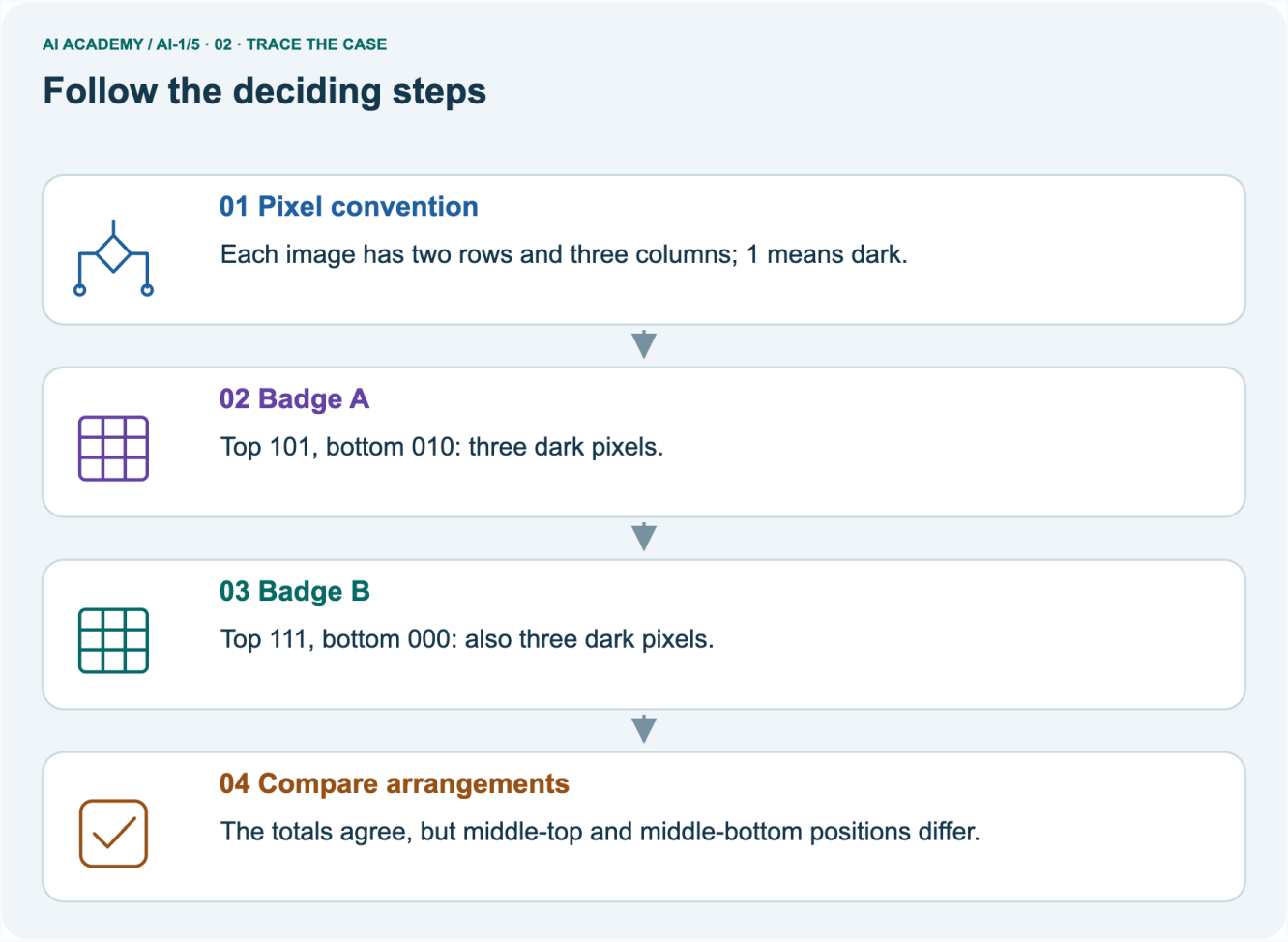
Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Compare the original arrangements



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace what a count keeps



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why equal counts are insufficient

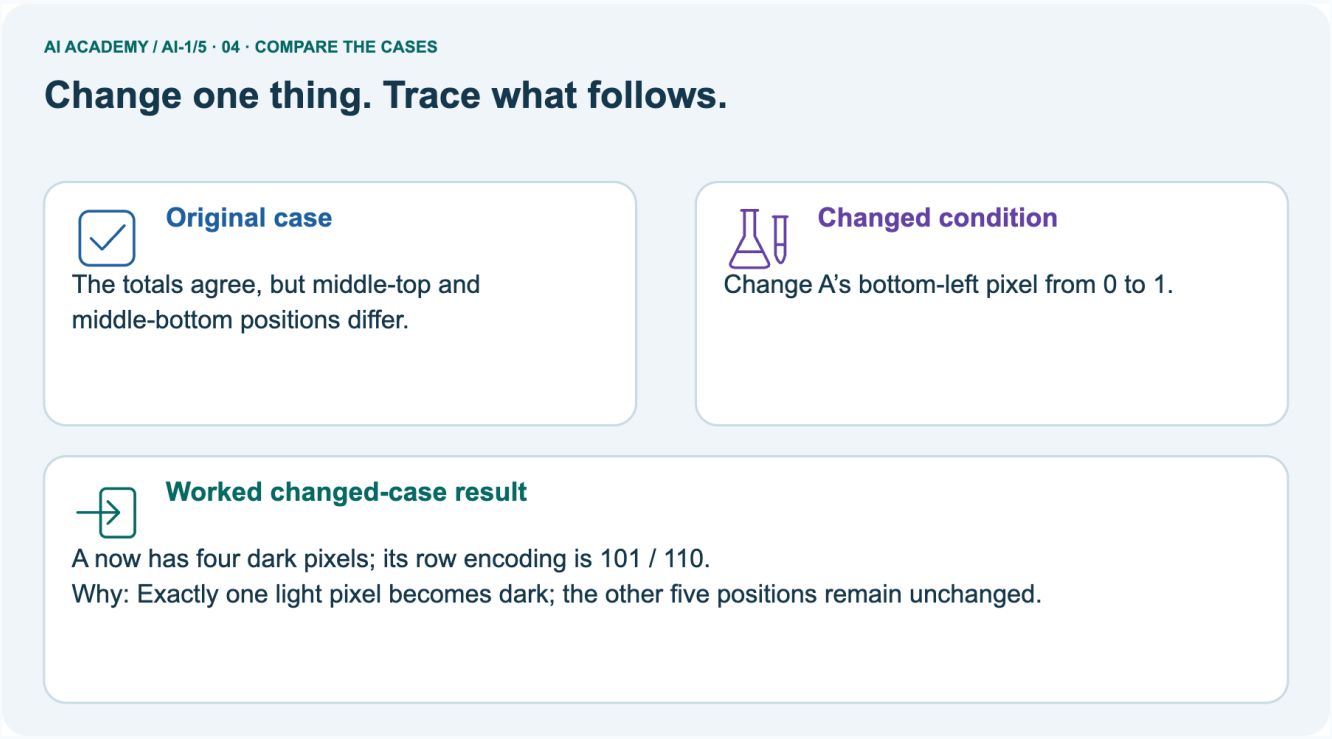
A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption.

Predict one changed pixel

- Before checking: change only Badge A's bottom-left pixel from 0 to 1. Write both complete new rows and the new dark count. Which positions stayed the same? Read your new code back as a picture.

▼ Check the changed badge



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the toy format's limits

This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

Read the position back

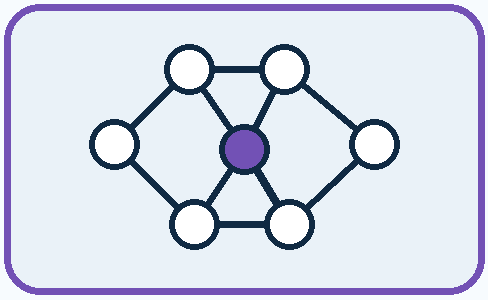
- Read the labelled picture: Find a position where the original badges differ, then read the changed A back as a picture.

▼ Earlier wording, alternative practice and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each **pixel** position.

STEP 1 — OBSERVE light measured at **grid** locations

STEP 2 — TRACE store and transform **pixel** values

STEP 3 — CONCLUDE a displayed, resized, blurred, or classified image

MODEL REASONING Each **grid** position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions.

The computer transforms numbers; the human viewer recognizes the shape.

DO NOT OVERCLAIM A **pixel** is a measurement at a location, not a tiny copy of the object.

What new **evidence** could change this answer?

A closer explanation

Suppose we discard the row records and retain only black **pixel** count=5. That count matches the cross, but it also matches rows 1,1,1 then 1,1,0 then 0,0,0. The second arrangement fills the top row and two positions below it.

Both have five black **pixel**s and nine **pixel**s in total, yet they are different images.

Stored information	What it establishes	What remains unknown
Width 3 and height 3	There are 9 positions	The value at each position

Stored information	What it establishes	What remains unknown
Black pixel count 5	Five positions are black under this code	Which five positions
Dimensions and all three ordered rows	The full binary grid under this convention	What a viewer or model will call the picture

A **pixel** count can still be a useful feature for some tasks. For example, it can separate empty **grid**s from **grid**s with at least one mark. But it cannot separate every shape.

To judge a feature, ask whether it retains the distinctions required by the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Pictures Become Numbers

The facts we will use

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/5

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Pixel convention

Each image has two rows and three columns; 1 means dark.



2. Badge A

Top 101, bottom 010: three dark pixels.



3. Badge B

Top 111, bottom 000: also three dark pixels.



4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Pixel convention	Each image has two rows and three columns; 1 means dark.
2. Badge A	Top 101, bottom 010: three dark pixels.
3. Badge B	Top 111, bottom 000: also three dark pixels.
4. Compare arrangements	The totals agree, but middle-top and middle-bottom positions differ.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Change A's bottom-left **pixel** from 0 to 1. Predict the result before checking the explanation. What remains the same?

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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MAIN LESSON · SECTION 7

Do not confuse more pixels with better evidence

> A reconstruction, an interpretation and an instruction make different claims.

More positions do not guarantee a useful image

Higher **resolution** offers more positions for recording detail, but it does not guarantee a useful image. A larger blurry image can still hide a feature, and a cropped image can omit it entirely.

A model's label also depends on what it was fitted to distinguish and the conditions represented in its examples.

Separate three levels of claim

Keep claims at the right stage. These nine values encode this **grid** is a reconstruction claim. This **grid** depicts a plus sign is an **interpretation**. This sign tells a person what to do adds context and authority.

One does not automatically establish the others.

Repair the tempting claims

- CLAIM A The computer stores a real cat inside the image file.
- CLAIM B More **pixel**s always guarantee a correct classification.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **Can two different three by three images have the same black-pixel count? Explain with a specific arrangement change**

▼ **Earlier wording, alternative practice and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Black/white icon	0 and 1 values	two possible brightness levels
Gray image	one brightness number per pixel	many light levels
Color image	red, green, blue values	three channels
Blur	nearby values are mixed	detail is reduced

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: reconstruct, read back and diagnose

> Use a round trip from code to drawing to code to locate a mismatch.

Draw the two-by-three practice grid

Use width 3, height 2 and rows 1,0,1 then 0,1,0. Draw two rows of three squares and fill exactly the positions marked 1. Check by reading your drawing back into numbers.

A round trip from code to drawing to code can reveal a swapped row or a missed **pixel**.

> **Draw the two by three image, write its black-pixel count and explain why saying only three black pixels would be insufficient**

Now restore a missing dimension

1. Your W2 message is 1,0,0,1,1,0. Show it as one row of width 6, then as two rows of width 3. In both versions keep the same value order, with 1 black and 0 white. Explain which missing field would tell a receiver which shape to use.

2. Use workbook W2 for the connected example “Images as numerical **grid**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Inspect the input before changing a classifier

If an image classifier mislabels a picture, first inspect the **input** representation. Was the width correct? Were the rows reversed? Did the program use the intended colour convention?

A later classifier cannot reliably repair every mistake in how its **input** was prepared.

Explain and exchange roles

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Does 1 always mean black?

- Use the worked case for Images as numerical **grid**s. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check the badge position

Find a position where the original badges differ, then read the changed A back as a picture.

> **My explanation using a pictured step and the given facts**

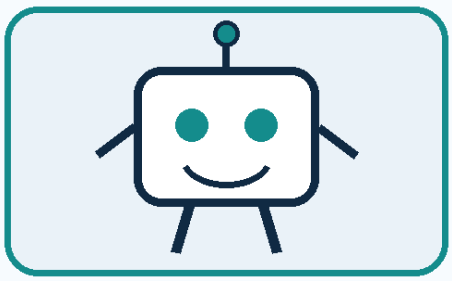
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternative practice and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A digital image is a **grid** of measured color values.

Use a fresh example to explain **pixel**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: More **pixel**s always guarantee a correct classification.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Preserve the missing field

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate: decode, then apply a toy blur

> Use an exact transformation so you can compare a prediction with the actual paper calculation.

Your complete decoding instructions

Use width 2 and height 2, with rows 0,1 / 1,0. Here 1 means black and 0 white. Predict the drawing, then decode it on paper and read it back. Keep the numeric grid separate from any name you give its pattern.

Predict and apply the averaging rule

1. For a declared toy blur, add the four values, divide by 4 and replace every position by that average. On this extended scale, 0.5 is halfway grey. Predict the result before calculating. Then try rows 1,0 / 0,1. Does this blur preserve which diagonal was dark? This is one fully specified averaging rule, not a claim about every real image-blurring method.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the original, result and limit

Case	Prediction	Observed result	What it means
Normal			
Edge			

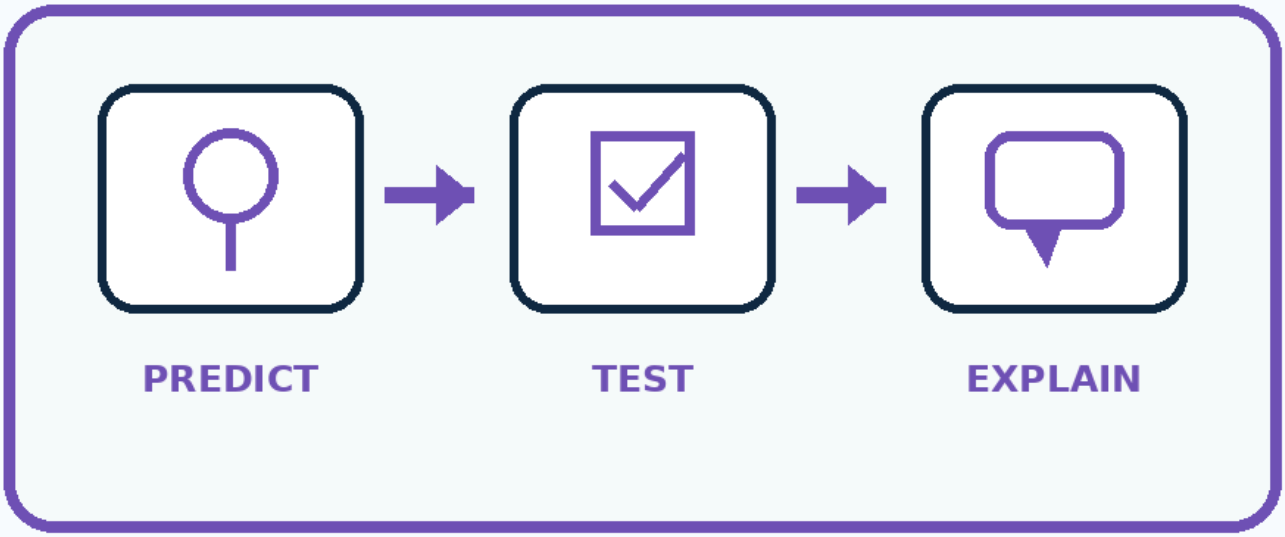
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternative practice and picture transcripts

EXPLORER LAB

Pixel Picture Decoder

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Decode a tiny **pixel** picture and predict a blurred version.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: create two new equal-count images

> Keep complete row encodings and explain what their shared count loses.

The new case and its code

For this new case, use 1 for black and 0 for white. Read each row left to right, with the top row first. Width 4 and height 2 means two rows of four values. Choose your own two arrangements.

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Design a new counterexample

Week 5 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left **pixel** to light. Which number changes?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the spatial clues your sorter needs

> Choose measurements for the task; do not assume a pixel total describes a shape.

This week’s design decision

THIS WEEK'S CHECKPOINT Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

Plan the representation and checks

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep the sensing boundary

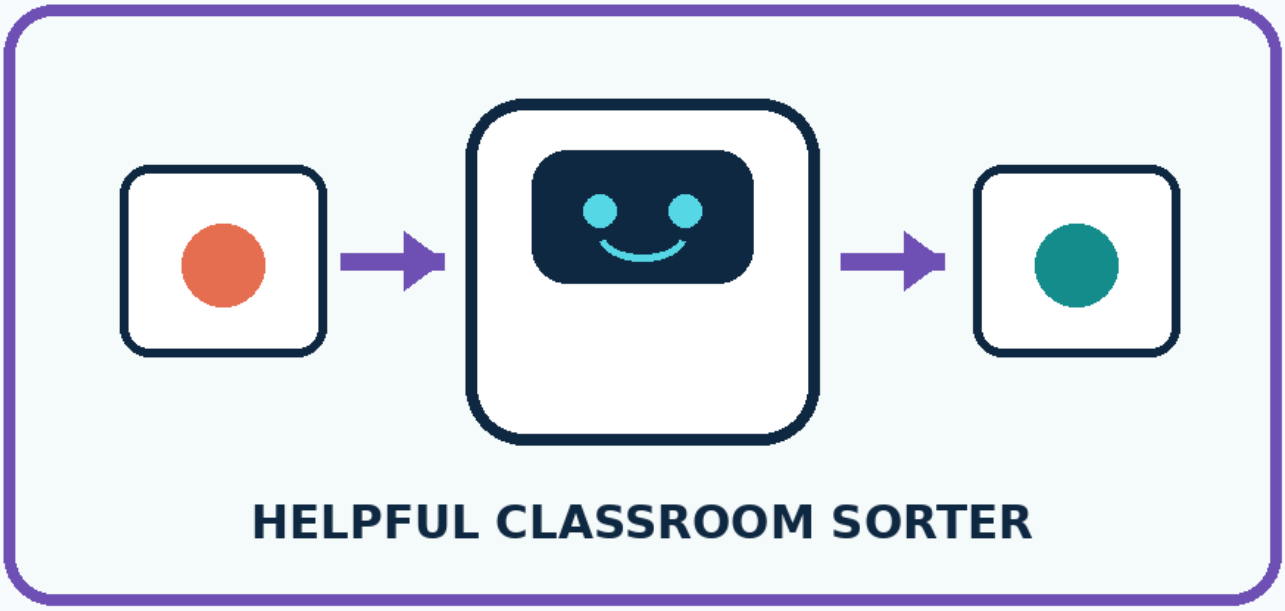
RESPONSIBLE DESIGN A **pixel** is a measurement at a location, not a tiny copy of the object.

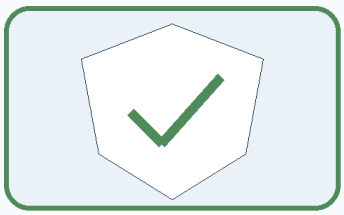
▼ Earlier wording, alternative practice and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain what the numbers preserve

> Close your notes and distinguish the encoded positions from the meaning assigned to a picture.

Your exit explanation

- Close your notes. Explain this goal in your own words: Encode and reconstruct a small image using an explicit **grid** convention

> **My deciding step, evidence and one remaining question**

Choose support and a later return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your image-representation word guide**

REFERENCE · SECTION 5

Your image-representation word guide

> Use each word to explain what a particular record preserves.

Pixels, grids, channels and resolution

Word	Meaning	My example
pixel	one picture element in a grid	
grid	rows and columns that locate values	
channel	one measured color component	
resolution	number of pixels used to represent an image	

Encoding, arrangement and interpretation

Word	Meaning
Encoding	A convention for representing information with symbols or numbers
Spatial arrangement	Where elements are located relative to one another
Interpretation	A meaning or category assigned to a representation using a rule, model or human judgment

Repair the claim

- CLAIM TO REPAIR The computer stores a real cat inside the image file.
- Use one vocabulary word and one piece of **evidence** to repair it:

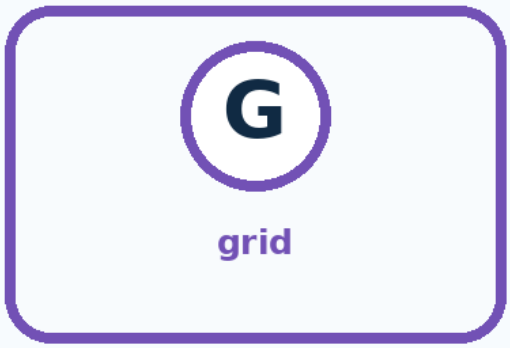
▼ Earlier wording, alternative practice and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a stronger image challenge**

REFERENCE · SECTION 10

Choose a stronger image challenge

> Choose one step that tests how well your representation supports a question.

Pick a useful challenge

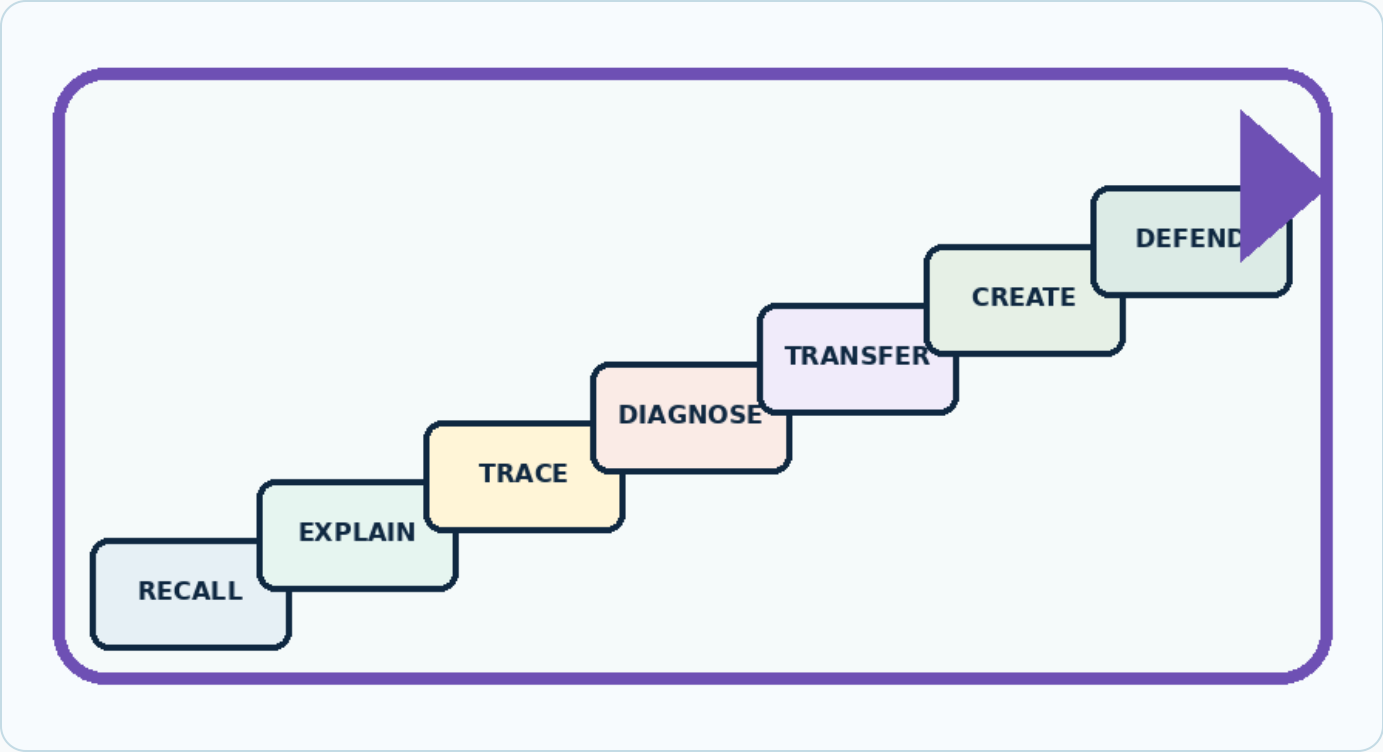
- RECALL Define **pixel**.
- EXPLAIN A digital image is a **grid** of measured color values.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The computer stores a real cat inside the image file.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternative practice and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check an image explanation**

REFERENCE · SECTION 12

Reference: check an image explanation

> Use after practice. The new two-by-four counterexample is this week's selected independent attempt.

Check the representation and limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

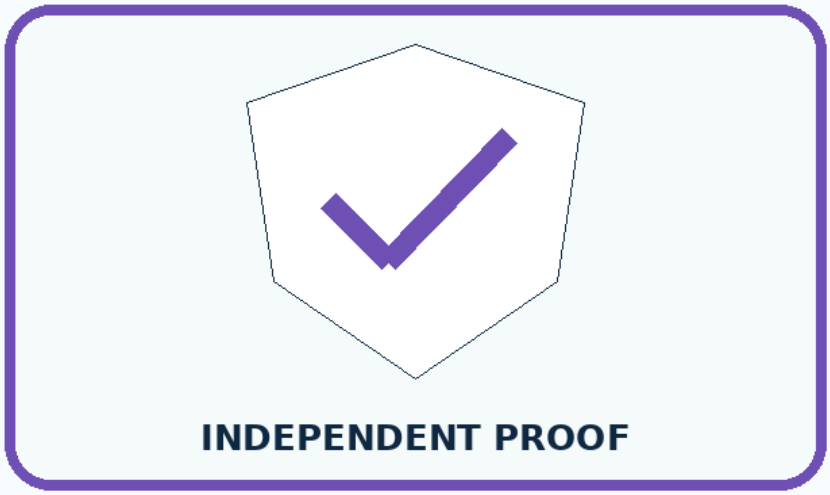
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternative practice and picture transcripts

MASTERY

Checklist for an image-representation explanation

Use this checklist after practice. The new two-by-four equal-count pair is this week's selected independent attempt.



EXPLANATION CHECK State dimensions, reading order and the colour convention; reconstruct the values and explain which spatial information a count discards.

Independent **evidence**:

SELF-CHECK Each **grid** position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions.

The computer transforms numbers; the human viewer recognizes the shape.

NEXT BRIDGE Carry today's **evidence** habit into: Sound Becomes a Signal.

▼ **Extra practice: change one square**

REFERENCE · SECTION 13

Extra practice: change one square

> Draw the small image before opening the worked explanation.

Read the case and requested change

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

Evidence or stage	Value or result
Top left	0
Top right	1
Bottom left	1
Bottom right	0

▼ Earlier wording, alternative practice and picture transcripts

Week 5 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A picture represented by four numbers

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

▼ Reference: original image lesson checklist

REFERENCE · SECTION 15

Reference: original image lesson checklist

> Use this additional checklist when it helps you inspect your work.

Targets and evidence

By the end, I can	How I will show it
Encode and reconstruct a small image using an explicit grid convention	Explain the deciding step, show a trace or test, and state one limit.
Explain why pixel counts alone cannot recover an arrangement	Explain the deciding step, show a trace or test, and state one limit.
Distinguish numerical image data from a model's interpretation of that data	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternative practice and picture transcripts

Start with this question: Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

▼ Extra practice: image pictures and feedback

REFERENCE · SECTION 17

Extra practice: image pictures and feedback

> Attempt each question before opening its explanation. Keep this practice separate from the reserved new case.

First supported question

- Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.
- Which labelled step matches this detail? The totals agree, but middle-top and middle-bottom positions differ.
- 1. Compare arrangements
- 2. Badge B
- 3. **Pixel** convention
- 4. Badge A
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare arrangements The totals agree, but middle-top and middle-bottom positions differ. A **grid** preserves where each brightness belongs.

Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

Changed-pixel question

- Changed-case practice: Change A's bottom-left **pixel** from 0 to 1. Which conclusion is supported by the supplied facts?
- 1. A now has four dark **pixel**s; its row **encoding** is 101 / 110.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed pixel**

After your attempt, compare with the supported explanation: A now has four dark **pixel**s; its row **encoding** is 101 / 110. Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Repair and save

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternative practice and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

</details>

<details>
<summary>Reference: six-picture image journey</summary>

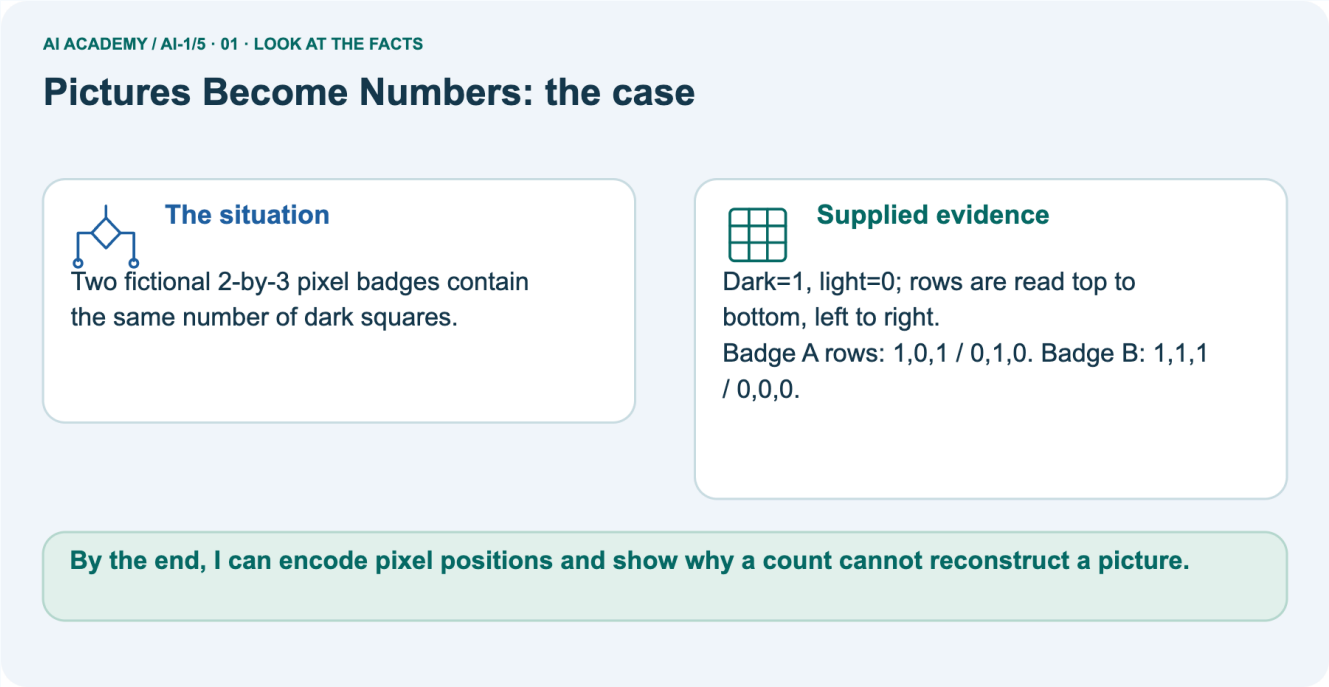
</details>

REFERENCE · SECTION 18

Reference: six-picture image journey

> Revisit the image that serves your current question; the main badge diagrams already sit beside the explanation.

01. Pictures Become Numbers: the case

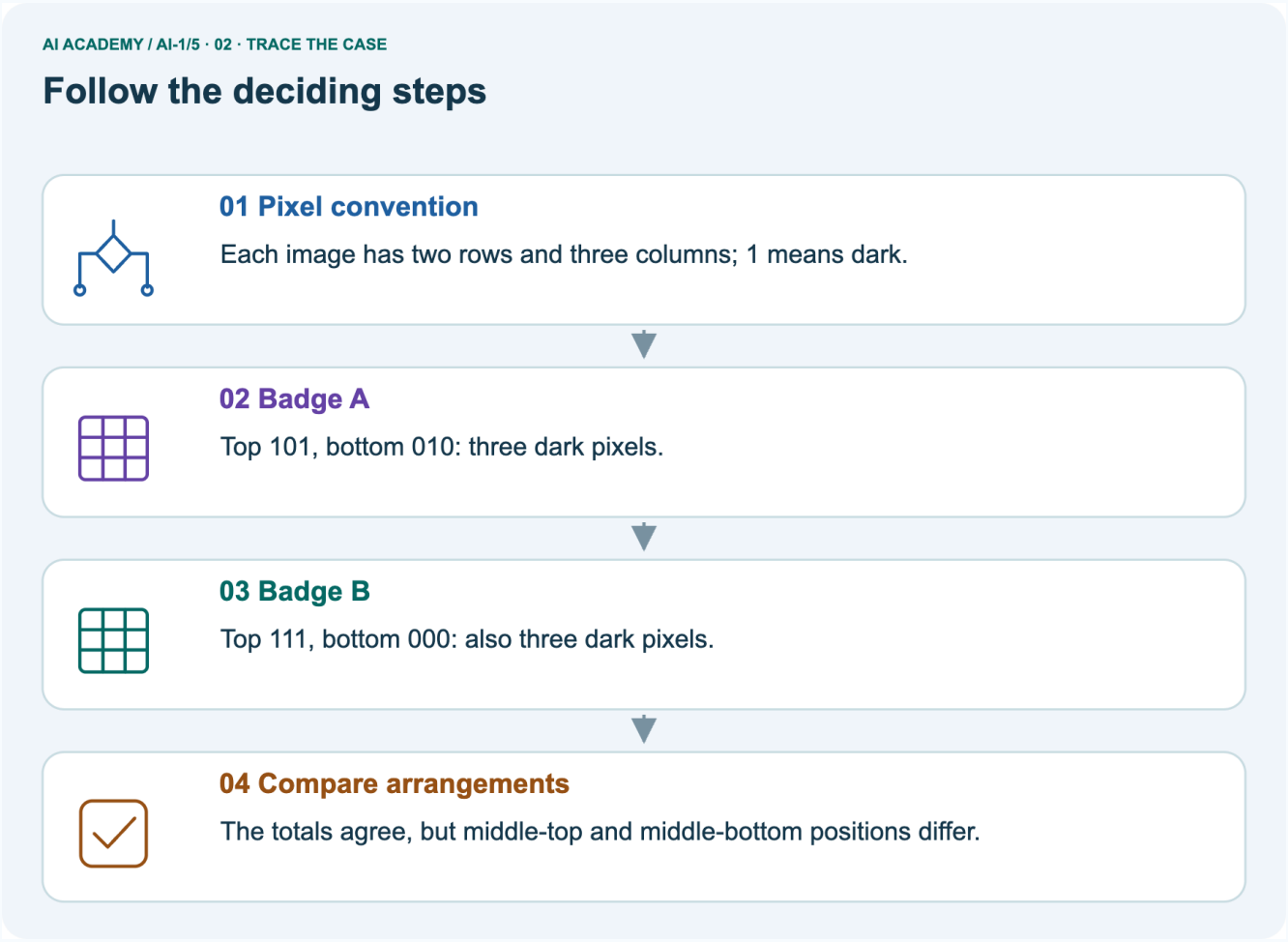


Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares. Supplied facts: Dark=1, light=0; rows are read top to bottom, left to right. Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

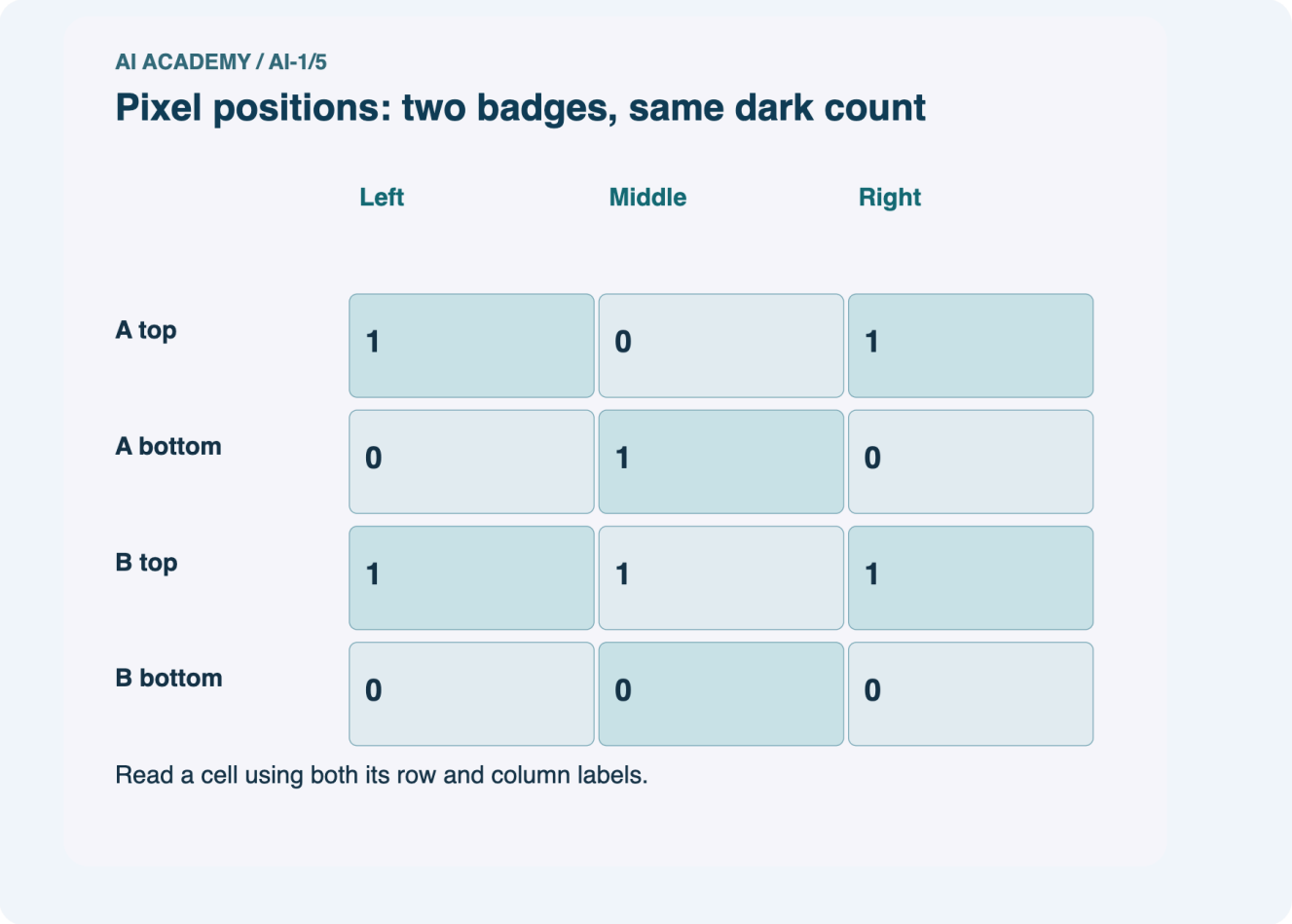


Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel convention: Each image has two rows and three columns; 1 means dark. → Badge A: Top 101, bottom 010: three dark **pixel**s. → Badge B: Top 111, bottom 000: also three dark **pixel**s. → Compare arrangements: The totals agree, but middle-top and middle-bottom positions differ.

Point to each numbered stage and explain which recorded **input** it uses.

03. Pixel positions: two badges, same dark count

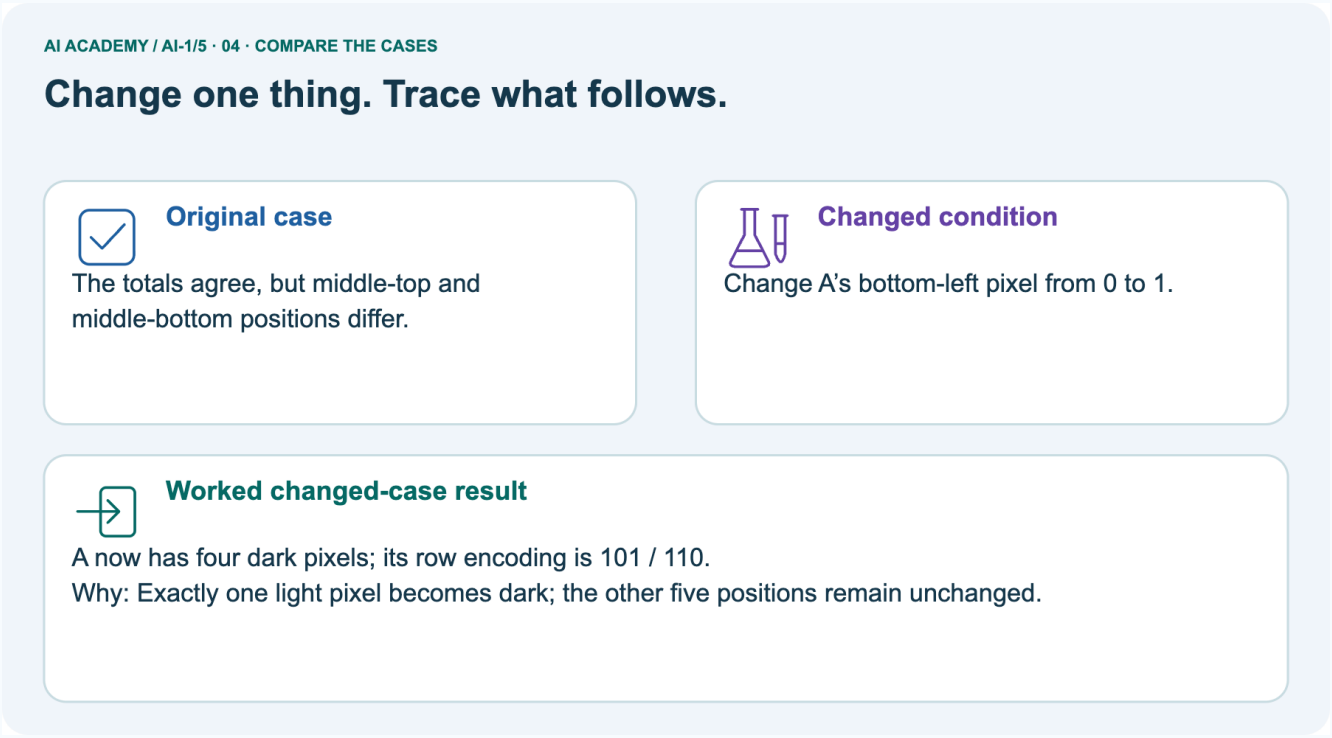


Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel positions: two badges, same dark count. A top: Left = 1, Middle = 0, Right = 1. A bottom: Left = 0, Middle = 1, Right = 0. B top: Left = 1, Middle = 1, Right = 1. B bottom: Left = 0, Middle = 0, Right = 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



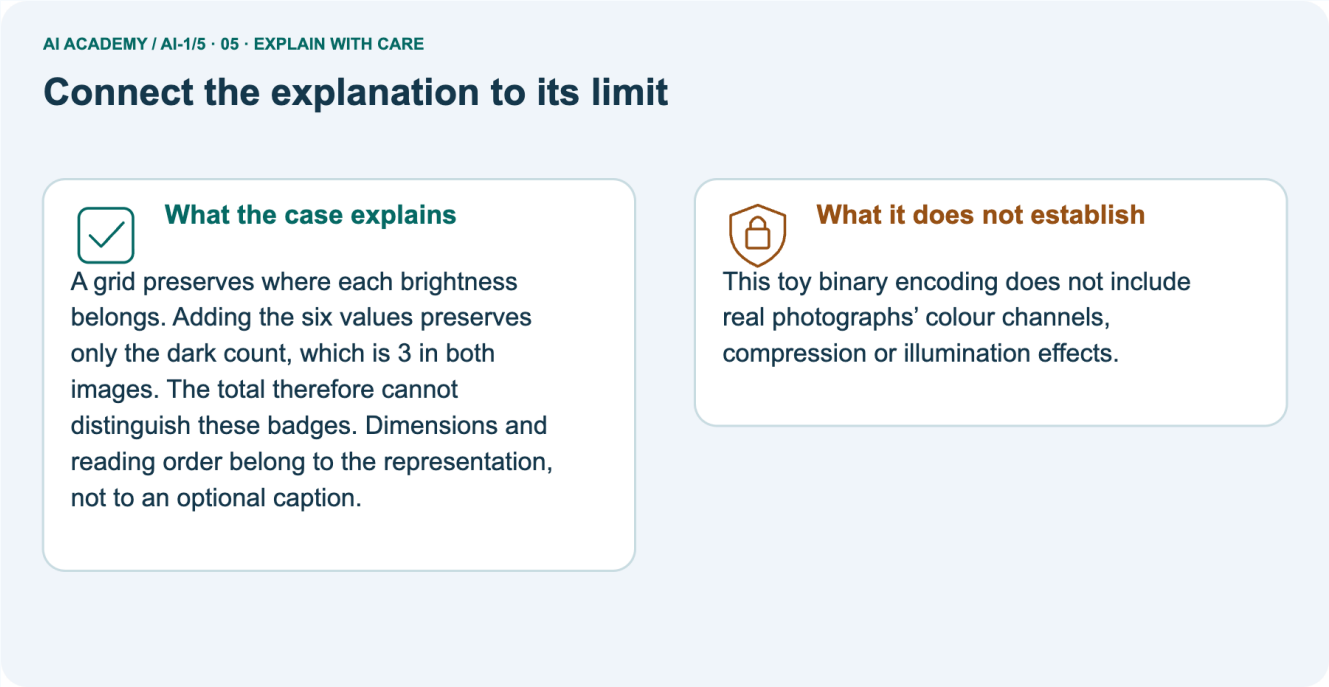
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The totals agree, but middle-top and middle-bottom positions differ. Changed condition: Change A's bottom-left **pixel** from 0 to 1. Expected result: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



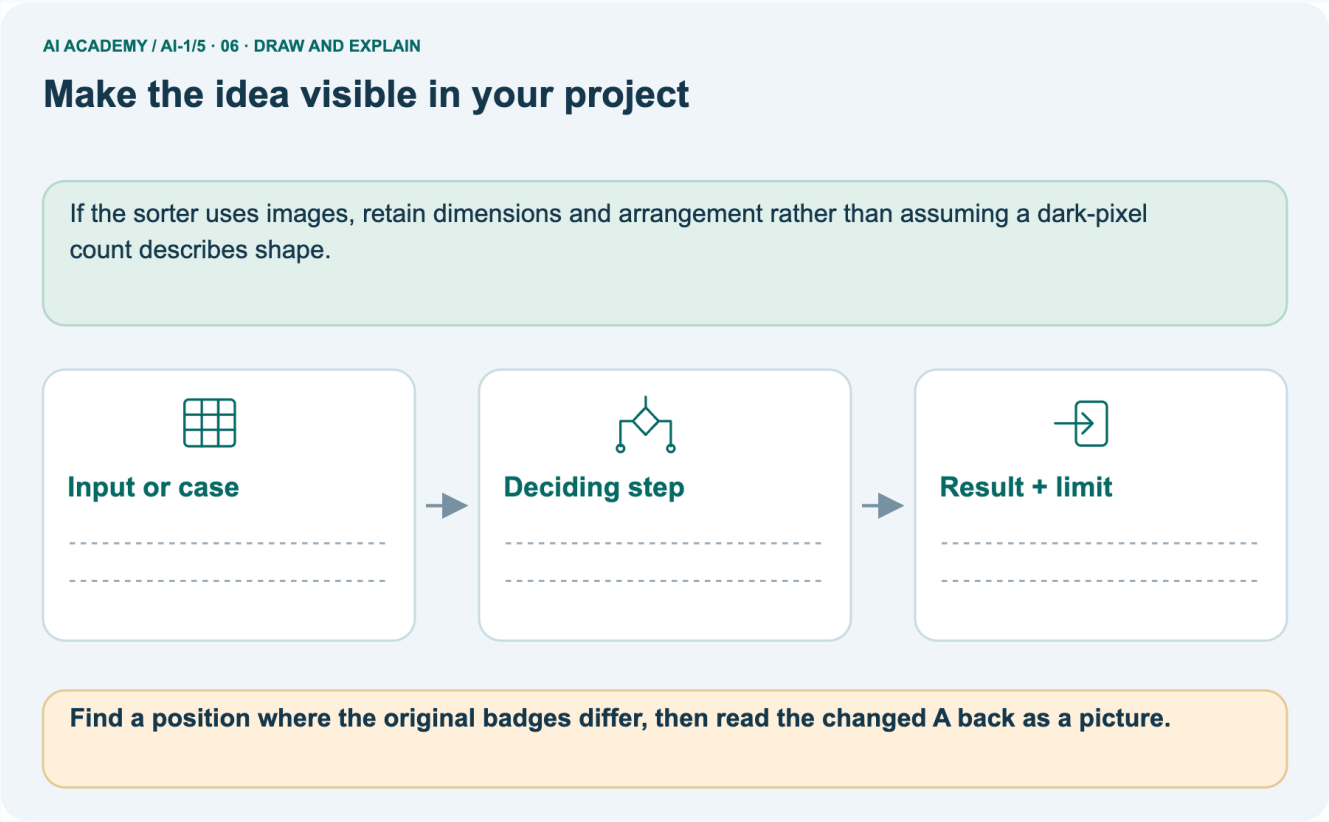
Follow the labelled picture with the matching explanation. Open the vector diagram

A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption. Limit: This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape. Draw your **input**, deciding step, result and limit.

Supported question: Find a position where the original badges differ, then read the changed A back as a picture.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternative practice and picture transcripts

01. Pictures Become Numbers: the case

02. Follow the deciding steps

03. Pixel positions: two badges, same dark count

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.